Linear Partial Differential Equation of First Order: (Lagrange's Method)

Linear Partial differential equation of first order are of the form

$$Pp + Qq = R \quad \text{—— (LE1)}$$

where P, Q and R are given functions of x, y and z (which do not involve p or q).

Our aim is to find a relation between x, y and z involving an arbitrary function. Equation (LE1) is frequently referred to as Lagrange's equation

Consider n independent variables $x_1, x_2, \ldots, x_n$ and one dependent variable f. and let

$$p_i = \frac{\partial f}{\partial x_i}; \quad i = 1, 2, \ldots, n.$$

Then if $X_1, X_2, \ldots, X_n$ and Y are functions of $x_1, x_2, \ldots x_n$ and f, equation (LE1) can be generalized as

$$X_1 p_1 + X_2 p_2 + \cdots + X_n p_n = Y \quad \text{—— (LE2)}$$

where $X_1, X_2, \ldots, X_n$ and Y are functions of n independent variables $x_1, x_2, \ldots, x_n$ and a dependent variable f.

Note:

For linearity of (LE1) p, q appear to the first degree only.

But P, Q and R may be any functions of x, y and z. For example,

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z^2 + x^2$$

Note :-

In ODE, z must also appear linearly.

Example: $x\frac{dz}{dx} = z^2 + x^2$ — non-linear equation.

Lagrange's method of solving the LPDE of order one namely

$Pp + Qq = R$

Theorem:

The general solution of the linear partial differential equation $Pp + Qq = R$ ——— ①

is $\phi(u, v) = 0$ ——— ②

where ϕ is an arbitrary function and

$u(x, y, z) = c_1$ and $v(x, y, z) = c_2$ ——— ③

form a solution of the equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \text{——— ④}$$

where P, Q, R are functions of x, y, z.

Proof:

From $\phi(u, v) = 0$

We get $Pp + Qq = R$

$$\left.\begin{aligned} \text{Where} \quad P &= \frac{\partial u}{\partial y}\cdot\frac{\partial v}{\partial z} - \frac{\partial u}{\partial z}\frac{\partial v}{\partial y} \\ Q &= \frac{\partial u}{\partial z}\frac{\partial v}{\partial x} - \frac{\partial u}{\partial x}\cdot\frac{\partial v}{\partial z} \\ R &= \frac{\partial u}{\partial x}\cdot\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\cdot\frac{\partial v}{\partial x} \end{aligned}\right\} \quad \text{——— ⑤}$$

which are equivalent to

$$\left.\begin{aligned} P\frac{\partial u}{\partial x} + Q\frac{\partial u}{\partial y} + R\frac{\partial u}{\partial z} &= 0 \\ P\frac{\partial v}{\partial x} + Q\frac{\partial v}{\partial y} + R\frac{\partial v}{\partial z} &= 0 \end{aligned}\right\} \quad \text{——— ⑥}$$

The main difficulty is to obtain arbitrary function φ of ②.

Let us consider the equations $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$ where c_1 and c_2 are arbitrary constants.

Differentiating these integrals we get,

$$\left.\begin{aligned} \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz &= 0 \\ \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz &= 0 \end{aligned}\right\} \text{——} ⑦$$

Solving the above equation ⑦ for dx, dy, dz, we get

$$\frac{dx}{\frac{\partial u}{\partial y}\cdot\frac{\partial v}{\partial z} - \frac{\partial u}{\partial z}\cdot\frac{\partial v}{\partial y}} = \frac{dy}{\frac{\partial u}{\partial x}\cdot\frac{\partial v}{\partial z} - \frac{\partial v}{\partial x}\cdot\frac{\partial u}{\partial z}} = \frac{dz}{\frac{\partial u}{\partial x}\cdot\frac{\partial v}{\partial y} - \frac{\partial v}{\partial x}\cdot\frac{\partial u}{\partial y}} \text{——} ⑧$$

Using the value of P, Q, R in ⑧, we obtain,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \text{——} ⑨$$

which are known as Lagrange's equations or Lagrange's auxiliary equations.

Note: If solutions of ⑨ are $u = a$, $v = b$, the general solution of the given PDE is

$$\varphi(u, v) = 0 \quad \text{or} \quad u = \varphi(v)$$

$$\varphi(a, b) = 0 \quad \text{or} \quad a = \varphi(b).$$

Corollary: If the equation is

$$P\frac{\partial z}{\partial x} + Q\frac{\partial z}{\partial y} + R\frac{\partial z}{\partial t} = S \text{——} (*)$$

having three independent variables x, y, t. Then the Lagrange's auxiliary equations are,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dt}{R} = \frac{dz}{S} \quad \text{———} \quad (**)$$

If $u = a$, $v = b$, $w = c$ are three independent solutions of $(**)$, then general solution of $(*)$ is

$$\phi(u, v, w) = 0$$

Thus this method can be generalized to any number of independent variables.

General Theorem:

If $u_i(x_1, x_2, \cdots, x_n, z) = c_i \; (i = 1, 2, \cdots, n)$ are independent solutions of the equations

$$\frac{dx_1}{P_1} = \frac{dx_2}{P_2} = \cdots = \frac{dx_n}{P_n} = \frac{dz}{R}$$

then the relation $\phi(u_1, u_2, \cdots, u_n) = 0$ in which the function ϕ is arbitrary, is a general solution of the linear partial differential equation

$$P_1 \frac{\partial z}{\partial x_1} + P_2 \frac{\partial z}{\partial x_2} + \cdots + P_n \frac{\partial z}{\partial x_n} = R$$

Proof:

First note that if the solutions of the equations

$$\frac{dx_1}{P_1} = \frac{dx_2}{P_2} = \cdots = \frac{dx_n}{P_n} = \frac{dz}{R} \quad \text{———} \quad (1)$$

are $u_i(x_1, x_2, \cdots, x_n, z) = c_i \; ; \; (i = 1, 2, \cdots, n)$ ——— (2)

then the n equations

$$\sum_{j=1}^{n} \frac{\partial u_i}{\partial x_j} dx_j + \frac{\partial u_i}{\partial z} dz = 0 \; ; \quad i = 1, 2, \cdots n \quad \text{———} \quad (3)$$

must be compatible with the equations (1). In other words, we must have

$$\sum_{j=1}^{n} P_j \frac{\partial u_i}{\partial x_j} + R\frac{\partial u_i}{\partial z} = 0 \quad \text{——— (4)}$$

Solving the set of n eqns (4) for P_i, we find that

$$\frac{P_i}{\dfrac{\partial(u_1,u_2,\dots,u_n)}{\partial(x_1,\dots,x_{i-1},z,x_{i+1},\dots,x_n)}} = \frac{R}{\dfrac{\partial(u_1,u_2,\dots,u_n)}{\partial(x_1,x_2,\dots,x_n)}} \; ; i=1,2,\dots n \quad \text{——— (5)}$$

where $\dfrac{\partial(u_1,u_2,\dots,u_n)}{\partial(x_1,x_2,\dots,x_n)}$ denotes the Jacobian.

$$\begin{vmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} & \cdots & \frac{\partial u_1}{\partial x_n} \\ \frac{\partial u_2}{\partial x_1} & \frac{\partial u_2}{\partial x_2} & & \frac{\partial u_2}{\partial x_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial u_n}{\partial x_1} & \frac{\partial u_n}{\partial x_2} & & \frac{\partial u_n}{\partial x_n} \end{vmatrix}$$

Consider now the relation

$$\phi(u_1,u_2,\dots,u_n) = 0 \quad \text{——— (6)}$$

Differentiating it with respect to x_i, we obtain the equation

$$\sum_{j=1}^{n}\left(\frac{\partial\phi}{\partial u_j}\frac{\partial u_j}{\partial x_i} + \frac{\partial u_j}{\partial z}\cdot\frac{\partial z}{\partial x_i}\right) = 0$$

and there are n such equations, one for each value of i

Eliminating the n quantities $\frac{\partial\phi}{\partial u_1}, \dots, \frac{\partial\phi}{\partial u_n}$ from these equations, we obtain the relation

$$\frac{\partial(u_1,\cdots,u_n)}{\partial(x_1,\cdots,x_n)} + \sum_{j=1}^{n} \frac{\partial z}{\partial x_j} \cdot \frac{\partial(u_1,\cdots,u_{j-1},u_j,u_{j+1},\cdots,u_n)}{\partial(x_1,\cdots,x_{j-1},z,x_{j+1},\cdots,x_n)} = 0 \quad \text{--- (7)}$$

Substituting from equations (5) into the equation (7), we see that the function z defined by the relation (6) is a solution of the equation

$$P_1\frac{\partial z}{\partial x_1} + P_2\frac{\partial z}{\partial x_2} + \cdots + P_n\frac{\partial z}{\partial x_n} = R$$

as we desired to show.

Find the general integrals of the Linear PDE.

Rule 1:

class ✓ Problem 1: (example-1 of Sneddon)

$$x^2\frac{\partial z}{\partial x} + y^2\frac{\partial z}{\partial y} = (x+y)z$$ (may not be by rule 1)

Solution:

The given PDE is

$$x^2\frac{\partial z}{\partial x} + y^2\frac{\partial z}{\partial y} = (x+y)z \quad \text{--- (1)}$$

The integral surfaces of the equation (1) are generated by the integral curves of the equations

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{(x+y)z} \quad \text{--- (2)}$$

From the first two term of eqn (2), we get

$$\frac{dx}{x^2} = \frac{dy}{y^2}$$

or, $-\frac{1}{x} = -\frac{1}{y} + c_1$

$\Rightarrow -\frac{1}{x} + \frac{1}{y} = c_1$

$\therefore c_1 = \frac{x-y}{xy}$ ——— (3)

Again from (2), we get,

$$\frac{dx - dy}{x^2 - y^2} = \frac{dz}{(x+y) z}$$

or, $$\frac{dx - dy}{x - y} = \frac{dz}{z}$$

or, $$\log(x-y) = \log z + \log c_2$$

$$\Rightarrow \log(x-y) - \log z = \log c_2$$

$$\Rightarrow \log \frac{x-y}{z} = \log c_2$$

$$\therefore c_2 = \frac{x-y}{z} \quad \text{———} (4)$$

Combining (3) and (4) we get

$$\frac{xy c_1}{z} = c_2$$

$$\Rightarrow \frac{xy}{z} = \frac{c_2}{c_1}$$

$$\therefore \frac{xy}{z} = c_3 \quad \text{———} (5)$$

and that the curves given by these equations generated the surface

$$F\left(\frac{xy}{z}, \frac{x-y}{z}\right) = 0$$

where the function F is arbitrary.

Note: It should be observed that this surface can be expressed by equations such as

$$z = xy\, f\left(\frac{x-y}{z}\right)$$

or $$z = xy\, g\left(\frac{x-y}{xy}\right)$$

where f and g are atrbitrary .

Problem 2:

$$yzp + zxq = xy$$

Solution:

Given $yzp + zxq = xy$ ——— ①

Lagrange's auxiliary eqns for ① are

$$\frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy} \quad \text{——— ②}$$

Taking first two term from ② we get

$$\frac{dx}{yz} = \frac{dy}{zx}$$

$$\Rightarrow \frac{dx}{y} = \frac{dy}{x}$$

$$\Rightarrow x\,dx = y\,dy$$

$$\Rightarrow \frac{x^2}{2} = \frac{y^2}{2} + \frac{c_1}{2}$$

$$\therefore x^2 - y^2 = c_1 \quad \text{——— ③}$$

Again taking first and third term we get

$$\frac{dx}{yz} = \frac{dz}{xy}$$

$$\Rightarrow \frac{dx}{z} = \frac{dz}{x}$$

$$\Rightarrow x\,dx = z\,dz$$

$$\Rightarrow \frac{x^2}{2} = \frac{z^2}{2} + \frac{c_2}{2}$$

$$\therefore x^2 - z^2 = c_2 \quad \text{——— ④}$$

③ and ④ are two independent solution of ②.

Hence the required general solution is

$$\phi\,(x^2 - y^2,\ x^2 - z^2) = 0$$

where ϕ is an arbitrary function.

Problem 3:
$$y^2p - xyq = x(z-2y)$$

Solution:
Given $y^2p - xyq = x(z-2y)$ —— ①

Here Lagrange's auxiliary eqns are

$$\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z-2y)} \quad —— ②$$

Taking the first two functions of ②, we get

$$\frac{dx}{y^2} = \frac{dy}{-xy}$$

$$\Rightarrow \frac{dx}{y} = \frac{dy}{-x}$$

$$\Rightarrow -xdx = ydy$$

$$\Rightarrow -\frac{x^2}{2} = \frac{y^2}{2} + \frac{c}{2}$$

$$\therefore x^2+y^2 = C_1$$

Now taking the second and 3rd term of ②, we get

$$\frac{dy}{-xy} = \frac{dz}{x(z-2y)}$$

$$\Rightarrow \frac{dy}{-y} = \frac{dz}{z-2y}$$

$$\Rightarrow \frac{dz}{dy} = \frac{z-2y}{-y}$$

$$\Rightarrow \frac{dz}{dy} = -\frac{1}{y}z + 2$$

$$\Rightarrow \frac{dz}{dy} + \frac{1}{y}z = 2$$

which is linear in z and y.

~~$\frac{dz}{dy} = \frac{z-2y}{y}$~~

~~taking, $z = vy$~~

~~$\Rightarrow \frac{dz}{dy} = v + y\frac{dv}{dy}$~~

~~$\therefore v + y\frac{dv}{dy} = \frac{v-2}{1}$~~

~~$\Rightarrow dv = -\frac{2dy}{y}$~~

~~$\Rightarrow v = -2\log y + c$~~

~~$\Rightarrow \frac{z}{y} + 2\log y = c$~~ ?

Its $I.F = e^{\int \frac{1}{y} dy} = e^{\log y} = y$

Multiplying by y we get,

$$y \frac{dz}{dy} + z = 2y$$

$$\Rightarrow d[yz] = 2y$$

$$\Rightarrow yz = y^2 + c_2$$

$$\therefore yz - y^2 = c_2$$

Hence $\varphi(x^2+y^2, zy-y^2) = 0$ is the desired general solution, where φ is an arbitrary function.

Problem 4: $p \tan x + q \tan y = \tan z$

Solution: Given $p \tan x + q \tan y = \tan z$ ——— ①

Here Lagrange's equations are

$$\frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z} \quad \text{——— ②}$$

Taking first two term of ②, we get,

$$\frac{dx}{\tan x} = \frac{dy}{\tan y}$$

$$\Rightarrow \cot x \, dx = \cot y \, dy$$

$$\Rightarrow \log \sin x = \log \sin y + \log c_1$$

$$\Rightarrow \log \sin x - \log \sin y = \log c_1$$

$$\Rightarrow \log \frac{\sin x}{\sin y} = \log c_1$$

$$\frac{\sin x}{\sin y} = c_1$$

Taking last two term of ② we get,

$$\frac{dy}{\tan y} = \frac{dz}{\tan z}$$

$$\Rightarrow \cot y\, dy = \cot z\, dz$$

$$\Rightarrow \log \sin y = \log \sin z + \log c_2$$

$$\Rightarrow \log \frac{\sin y}{\sin z} = \log c_2$$

$$\therefore \frac{\sin y}{\sin z} = c_2$$

The required general solution is,

$$\phi\left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z} \right) = 0$$

Problem 5: $y^2zp + x^2zq = xy^2$.

Solution: Given $y^2zp + x^2zq = xy^2$ —— ①

Lagrange's auxiliary equations are

$$\frac{dx}{y^2z} = \frac{dy}{x^2z} = \frac{dz}{xy^2} \quad \text{——} ②$$

Taking the first two functions, we get

$$\frac{dx}{y^2z} = \frac{dy}{x^2z}$$

$$\Rightarrow \frac{dx}{y^2} = \frac{dy}{x^2}$$

$$\Rightarrow x^2 dx = y^2 dy$$

$$\Rightarrow \frac{x^3}{3} = \frac{y^3}{3} + \frac{c}{3}$$

$$\therefore x^3 - y^3 = c_1 \quad \text{——} ③$$

Next, taking the first and third functions, we get

$$\frac{dx}{y^2z} = \frac{dz}{xy^2}$$

$$\Rightarrow x\,dx = z\,dz$$

$$\Rightarrow \frac{x^2}{2} = \frac{z^2}{2} + \frac{c}{2}$$

$$\therefore x^2 - z^2 = c_2 \quad \text{——— (4)}$$

Since $x^3 - y^3$ and $x^2 - z^2$ are independent, the required solution is given by $\phi(x^3 - y^3, x^2 - z^2) = 0$ where ϕ is an arbitrary function.

Rule II :

Problem 1 : $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = x + y + z$

Solution : Given $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = x + y + z$

$$\Rightarrow p + q = x + y + z \quad \text{——— (1)}$$

Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{1} = \frac{dz}{x+y+z} \quad \text{——— (2)}$$

Taking the first two term of (2), we get

$$dx = dy$$

$$\Rightarrow x = y + c_1$$

$$\Rightarrow x - y = c_1$$

Taking the 2nd and 3rd term of (2), we get

$$dy = \frac{dz}{x+y+z}$$

$$\Rightarrow \frac{dz}{dy} = x + y + z$$

$$\Rightarrow \frac{dz}{dy} = y + c_1 + y + z$$

$$\Rightarrow \frac{dz}{dy} - z = 2y + c_1 \quad \text{——— (3)}$$

which is linear in z and y.

Its $I.F = e^{\int -dy} = e^{-y}$

Multiplying by e^{-y}, we get

$$e^{-y}\frac{dz}{dy} - e^{-y}z = c_1 e^{-y} + 2ye^{-y}$$

$$\Rightarrow d[e^{-y}z] = e^{-y}(c_1 + 2y)$$

$$\Rightarrow ze^{-y} = \int e^{-y}(c_1 + 2y)\,dy$$

$$= \int c_1 e^{-y}dy + 2\int ye^{-y}dy$$

$$= -c_1 e^{-y} + 2\left[-ye^{-y} - \int -e^{-y}dy\right]$$

$$= -c_1 e^{-y} + 2[-ye^{-y} - e^{-y}] + c_2$$

$$\Rightarrow ze^{-y} + 2ye^{-y} + 2e^{-y} + c_1 e^{-y} = c_2$$

$$\Rightarrow e^{-y}(z + 2y + 2 + x - y) = c_2$$

$$\Rightarrow e^{-y}(x + y + z + 2) = c_2$$

So the required general solution is

$$\varphi\left[x - y,\ e^{-y}(x+y+z+2)\right] = 0.$$

Problem 2: $z(p - q) = z^2 + (x+y)^2$

Solution: Given $zp - zq = z^2 + (x+y)^2$ ——— (1)

Lagrange's auxiliary eqns are

$$\frac{dx}{z} = \frac{dy}{-z} = \frac{dz}{z^2 + (x+y)^2} \quad \text{——— (2)}$$

Taking the first two function from ②, we get

$$\frac{dx}{z} = \frac{dy}{-z}$$

$\Rightarrow dx = -dy$

$\Rightarrow x = -y + c_1$

$\therefore x + y = c_1$ ——— ③

Taking the last two function in ② and using ③ we get,

$$\frac{dy}{-z} = \frac{dz}{z^2 + c_1^2}$$

$$\Rightarrow dy = -\frac{z\,dz}{z^2 + c_1^2}$$

$$\Rightarrow 2\,dy = -\frac{2z\,dz}{z^2 + c_1^2}$$

$\Rightarrow \log(z^2 + c_1^2) = -2y + \log c_2$

$\Rightarrow \log(z^2 + c_1^2) - \log c_2 = \log e^{-2y}$

$$\Rightarrow \log \frac{z^2 + c_1^2}{c_2} = \log e^{-2y}$$

$$\Rightarrow \frac{z^2 + c_1^2}{c_2} = e^{-2y}$$

$\Rightarrow e^{2y}(z^2 + c_1^2) = c_2$

$\Rightarrow e^{2y}[z^2 + (x+y)^2] = c_2$ ——— ④ ;[by ③]

From ③ and ④, the required general soluti is $\phi[x+y,\ e^{2y}\{z^2 + (x+y)^2\}] = 0$.

<u>Problem 3:</u>

$p - 2q = 3x^2 \sin(y + 2x)$

<u>Solution:</u> Given $p - 2q = 3x^2 \sin(y+2x)$ ——— ①